

Lyre Build Packet Print Packet

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Readiness Boundary

This is a prototype-planning scaffold for a compact 10-string plucked lyre. It supports design review, sourcing review, string schedule review, and first CAD layout. It is not build-ready and does not claim measured tuning data, reviewed CAD/DXF authority, verified sourcing, or a shop-validated frame under string load.

File Map

File	Purpose
design.md	Design narrative and readiness boundary.
string-schedule.csv	String target schedule and unresolved gauge/tension fields.
bridge-soundboard-load-register.csv	Bridge, soundboard, tuning pin, yoke, and joinery load gates.
visual-output-register.csv	Visual authority register.
bom.csv	Starter bill of materials.
sourcing.csv	Supplier/search tracker.
cut-list.csv	Stock and cut plan.
drawing-brief.md	Drawing authority brief.
cad/README.md	CAD skeleton notes.
drawings/README.md	Drawing queue notes.
assembly-manual.md	Shop-facing sequence.
validation.csv	Measurement and validation gates.
supplier-rfq.md	Supplier RFQ template.
visual-bom-brief.md	Visual BOM brief.
photo-shotlist.md	Documentation shot list.
risks.md	Risk register.
README.md	Repository overview.
wolfram-starter.wl	Local Wolfram starter.

design.md

Design narrative and readiness boundary.

Lyre Design

Intent

Create a small bench-buildable lyre packet that can move from concept to a first shop prototype without hiding the important string-load unknowns. The baseline is a compact 10-string plucked instrument with a flat soundboard, two arms, a crossbar/yoke, and individual tuning pins.

Readiness

Prototype-planning scaffold. This is suitable for design review, sourcing review, and first CAD layout. It is not ready for unattended fabrication.

Governing Model

Strings use the Mersenne-Taylor relationship:

$$f = (1 / (2 L)) * \text{sqrt}(T / \mu)$$
$$\mu = \text{density} * \text{pi} * (\text{diameter} / 2)^2$$

Use A4 = 440 Hz equal temperament for targets. Treat all string tensions as unknown until the actual string material, gauge, and measured or supplier density/breaking strength are recorded in string-schedule.csv.

Baseline Assumptions

Parameter	Baseline	Notes
String count	10	Diatonic C4-D4-E4-F4-G4-A4-B4-C5-D5-E5
Scale length	21 in nominal	May be fanned 18-23 in for ergonomics and tension
String material	Nylon or fluorocarbon	Gut is possible but not the first prototype default
Soundboard	0.125-0.160 in spruce, cedar, or birch ply	Final thickness requires tap and deflection checks
Frame stock	0.75 in hardwood	Grain must run continuously through arms where possible
Tuning hardware	Zither pins	Pilot holes and pin torque need validation
Target total tension	80-120 lbf	Design frame with margin before stringing

String Schedule Boundary

string-schedule.csv names the C4-E5 target set and holds the fields that must be resolved before the frame is treated as structurally reviewed:

- gauge_or_diameter_in
- linear_density_source
- estimated_tension_lbf
- percent_breaking
- risk_flags

Rows with status=TBD or blank tension fields are placeholders. They are useful for layout discussion, but they are not a stringing recipe or a load proof.

Bridge, Soundboard, And Joinery Gates

bridge-soundboard-load-register.csv is the load-path checklist for the first prototype. It keeps bridge downbearing, soundboard deflection, pin hold, crossbar/yoke grain direction, and arm-to-yoke joinery visible as separate gates. Do not drill final pin holes, cut final bridge/nut slots, or publish CNC toolpaths until these rows have measured or reviewed evidence.

Tension Safety Notes

- Keep the first prototype below 70 percent of estimated breaking load for all treble monofilament strings.
- Confirm the crossbar/yoke is not loaded in short grain.
- Bring strings to pitch gradually, alternating across the frame.
- Stop the build if the soundboard crowns, the yoke twists, or tuning pins creep under static load.

Known Gaps

- No measured string-gauge schedule yet.
- No CAD frame deflection check yet.
- No bridge downbearing measurement yet.
- No joinery load-path drawing or proof-load coupon yet.
- No supplier price or stock verification yet.
- No prototype tuning data.

string-schedule.csv

String target schedule and unresolved gauge/tension fields.

st ri n g _i d	tar get _n ote	tar get _h z	scale _len gth_i n	material _family	gauge_ or_diam eter_in	linear_d ensity_s ource	estimat ed_tens ion_lbf	perce nt_br eakin g	retu ning_ fact or	s t a t u s	risk_flags	next_ actio n	not es
L Y R- S 0 1	C4	26 1. 63	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.00 0	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	Ba ss- sid e first arti cle row ; do not drill fina l bri dge /nut slot s fr om this pla ceh old er.
L Y R- S 0 2	D4	29 3. 66	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.12 2	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	

st ri n g _i d	tar get _n ote	ta rg et _h z	scale _len gth_i n	material _family	gauge_ or_diam eter_in	linear_d ensity_s ource	estimat ed_tens ion_lbf	perce nt_br eakin g	retu ning_ fact or	s t a t u s	risk_flags	next_ action	not es
L Y R- S 0 3	E4	32 9. 63	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.26 0	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	
L Y R- S 0 4	F4	34 9. 23	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.33 5	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	
L Y R- S 0 5	G4	39 2. 00	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.49 8	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	

st ri n g _i d	tar get _n ote	ta rg et _h z	scale _len gth_i n	material _family	gauge_ or_diam eter_in	linear_d ensity_s ource	estimat ed_tens ion_lbf	perce nt_br eakin g	retu ning_ fact or	s t a t u s	risk_flags	next_ actio n	not es
L Y R- S 0 6	A4	44 0. 00	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.68 2	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	A4 ref ere nce row ; tu ner cali bra tion mu st be l ogg ed bef ore acc ept anc e.
L Y R- S 0 7	B4	49 3. 88	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			1.88 8	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	

st ri n g _i d	tar get _n ote	ta rg et _h z	scale _len gth_i n	material _family	gauge_ or_diam eter_in	linear_d ensity_s ource	estimat ed_tens ion_lbf	perce nt_br eakin g	retu ning_ fact or	s t a t u s	risk_flags	next_ actio n	not es
L Y R- S 0 8	C5	52 3. 25	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			2.00 0	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	
L Y R- S 0 9	D5	58 7. 33	21.0	nylon_o r_fluoro carbon	TBD	measur ement_r equired			2.24 5	T B D	unknown_ga uge;unknown _tension	Selec t sup plier string data or m easur e sa mple diam eter/ densi ty bef ore pi tch-u p	

string_id	target_note	target_hz	scale_length_in	material_family	gauge_or_diameter_in	linear_density_source	estimated_tension_lbf	percent_breaking	returning_factor	status	risk_flags	next_action	notes
LYRS10	E5	659.25	21.0	nylon_or_fluorocarbon	TBD	measurement_required			2.520	TBD	unknown_gauge;unknown_tension	Select supplier string data or measure sample diameter/density before pitch-up	Treble-side row; keep below 70 percent breaking load after gauge selection.

bridge-soundboard-load-register.csv

Bridge, soundboard, tuning pin, yoke, and joinery load gates.

check_id	area	artifact_or_member	claim_status	evidence_required	acceptance_gate	next_action	notes
LOAD-001	string_schedule	string-schedule.csv	measurement_required	supplier density/breaking strength or measured sample diameter and mass	each string has gauge, estimated_tension_lbf, percent_breaking, and risk flags resolved before full pitch-up	Choose a real string set or measure sample strings	Current rows are pitch targets and placeholders, not a final load table.
LOAD-010	total_frame_load	crossbar/yoke	measurement_required	summed string tensions from resolved schedule plus safety factor	total estimated load is recorded and reviewed against yoke grain direction and joint design	Update after string schedule is resolved	Do not drill the pin row until this check is complete.

check_id	area	artifact_or_member	claim_status	evidence_required	acceptance_gate	next_action	notes
LOAD-020	tuning_pin_hol d	tuning pin scrap block	measurement_r equied	scrap-block pilot-hole torque test and 24-hour creep observation	pins hold three retune cycles with <=5 cents drift per hour	Drill a scrap coupon with selected pin and wood species	Blocks trust in pitch-up data.
LOAD-030	bridge_downbe aring	bridge and nut	measurement_r equied	bridge height, break angle, slot radius, and downbe aring estimate	bridge/nut slots are polished and do not cut strings during pitch-up	Make a bridge/n ut coupon after string gauge selection	No final slot spacing until string gauges are chosen.
LOAD-040	soundboard_de flection	soundboard/t op	measurement_r equied	flatness before stringing, half-tension deflection, full-tension deflection, and 24-hour recheck	permanent movement is <=0.030 in and no crown/crack develops	Record deflection with str aightedge or dial indicator	Soundb oard thic kness remains a prototy pe assu mption.
LOAD-050	joinery_load_p ath	arms-to-yoke and body joints	measurement_r equied	full-size drawing or CAD notes showing long-grain load path, glue area, and pin edge distance	no loaded member depends on short grain; pin holes meet edge-distance gate	Add dra wing/CA D detail before cutting final frame	This is the main structura l review gate.
LOAD-060	prototype_pitch up	assembled first article	measurement_r equied	pitch-up log with before/after yoke distance, arm spread, bridge height, and tuning drift	no visible frame movement and pitch drift is documented after 24 hours	Fill valid ation.csv and phot o-shotlist evidence after first pitch-up	Prototyp e eviden ce required before r eadines s promot ion.

visual-output-register.csv

Visual authority register.

artif act_i d	path	artifact_ kind	role	authori ty	derive d_fro m	dimensi on_clai m	fabricatio n_require d	notes
IMG -001	images/P-Lyre-Harp-1200x1200-1_1080xinspiration.webp	photo_r eferenc e	photo_r eferenc e	referen ce_onl y		none	false	Inspiration /reference image only; does not control string scale, bridge pla cement, s oundboar d thickness, joinery, or toolpaths.
HTM L-00 1	explorer.html	render_ preview	story	concep t_only		none	false	Studio review surface for packet navigation ; not fabrication authority.
PDF -001	print-packet.pdf	render_ preview	story	concep t_only		none	false	Printable packet export for review; di mensions remain subject to string schedule and drawing authority gates.
PPT -001	capstone-deck.pptx	render_ preview	story	concep t_only		none	false	Presentati on export only; does not control fabrication geometry.

bom.csv

Starter bill of materials.

item	qty	unit	material_or_spec	estimated_cost_usd	notes
Frame hardwood blank	1	ea	maple, cherry, walnut, or similar clear hardwood, 0.75 in nominal	45	Avoid knots and short grain through arms

item	qty	unit	material_or_spec	estimated_cost_usd	notes
Soundboard panel	1	ea	spruce, cedar, or birch aircraft ply, 0.125-0.160 in	25	Prototype thickness to be validated by deflection and tone
Back or rim spacer stock	1	set	hardwood or plywood strips	15	Supports soundbox depth if boxed body is selected
Zither tuning pins	10	ea	0.197 in or compatible lyre/harp pin	18	Pilot-hole fit and torque must be tested on scrap
Bridge and nut stock	2	ea	hard maple, bone, or dense hardwood	8	Round contact points to reduce string breakage
String set	1	set	nylon or fluorocarbon monofilament gauges TBD	35	Gauge schedule must be calculated and tested
Adhesive	1	lot	wood glue suitable for instrument joints	8	Use fresh adhesive and full clamp coverage
Finish	1	lot	shellac, oil-varnish, or waterborne clear finish	15	Do not finish string contact surfaces until fit is final

sourcing.csv

Supplier/search tracker.

item	required_spec	search_terms	supplier_candidates	date_checked	unit_price_usd	lead_time	substitution_risk	notes
Zither pins	10 matching pins with compatible wrench and pilot recommendations	zither tuning pins harp lyre 0.197 inch	music hardware suppliers; luthier suppliers	TBD	TBD	TBD	Medium	Pin diameter and wood species affect holding torque

item	required_spec	search_terms	supplier_candidates	date_checked	unit_price_usd	lead_time	substitution_risk	notes
Monofilament strings	nylon or fluorocarbon gauges from calculated schedule	nylon monofilament harp strings fluorocarbon music string	harp string suppliers; fishing leader only if specs are documented	TBD	TBD	TBD	High	Density and breaking strength must be known
Soundboard stock	quarter-sawn softwood or stable thin ply	spruce soundboard blank cedar soundboard birch aircraft plywood	luthier suppliers; specialty plywood suppliers	TBD	TBD	TBD	Medium	Tone and deflection vary by material
Frame hardwood	clear straight-grain hardwood, 0.75 in nominal or thicker	instrument hardwood maple cherry walnut blank	local hardwood dealer; luthier tonewood supplier	TBD	TBD	TBD	Low	Avoid runout at arms
Bridge/nut stock	dense wear-resistant string contact material	hard maple bridge blank bone nut blank	luthier suppliers	TBD	TBD	TBD	Low	Must polish smooth to prevent string cuts

cut-list.csv

Stock and cut plan.

part	qty	rough_dimensions_in	final_dimensions_in	material	grain_or_orientation	operation	notes
Left arm	1	1.0 x 2.0 x 22.0	0.75 x profile x 21.0	clear hardwood	grain follows arm curve	bandsaw/CNC profile then rasp/sand	Leave extra at yoke joint until fitted
Right arm	1	1.0 x 2.0 x 22.0	0.75 x profile x 21.0	clear hardwood	grain follows arm curve	bandsaw/CNC profile then rasp/sand	Mirror left arm
Crossbar/yoke	1	1.0 x 1.5 x 16.0	0.75 x 1.0 x fitted length	clear hardwood	long grain end to end	drill pin row after layout verification	No short-grain load path

part	qty	rough_dimensions_in	final_dimensions_in	material	grain_or_orientation	operation	notes
Soundboard	1	0.160 x 13.0 x 18.0	0.125-0.160 x body profile	spruce/cedar /ply	long grain vertical	template trim and brace if needed	Final thickness TBD
Bridge	1	0.5 x 0.5 x 12.0	profile TBD	hard maple or bone	long grain along string row	shape, slot, polish	Confirm spacing before slots
Nut	1	0.375 x 0.5 x 12.0	profile TBD	hard maple or bone	long grain along string row	shape, slot, polish	Slots must match final gauges

drawing-brief.md

Drawing authority brief.

Lyre Drawing Brief

Required first drawings:

- Full-size front elevation with string fan, bridge, nut, yoke, and pin row.
- Side section through body/soundboard showing bridge height and downbearing.
- Crossbar/yoke detail with pin spacing, pilot diameter, and edge distance.
- Arm joint detail showing grain direction and glue surface.
- Soundboard bridge patch, brace or reinforcement zones, and no-cut/no-carve zones around loaded areas.
- Explicit datum from string-schedule.csv so bridge, nut, and pin row update

from the string paths rather than from decoration.

All drawings should label dimensions in inches, include metric equivalents for hardware, and mark TBD dimensions that depend on final string gauges. Drawing outputs remain draft review artifacts until the string schedule and load register have selected-string and prototype evidence.

cad/README.md

CAD skeleton notes.

CAD Starter Notes

Model the lyre as separate frame, soundboard, bridge, nut, and pin components. Expose parameters for string count, string spacing, nominal scale length, yoke width, soundboard thickness, and bridge height. Add construction planes for each string path so the bridge and pin row can be regenerated from the string table.

Minimum skeleton parameters:

- string_count
- scale_length_in
- string_spacing_bridge_in
- string_spacing_pin_row_in
- bridge_height_in
- soundboard_thickness_in
- pin_edge_distance_in
- yoke_section_height_in
- arm_section_width_in
- bridge_patch_length_in

The CAD model is a source scaffold until string-schedule.csv and bridge-soundboard-load-register.csv have enough evidence to declare a reviewed fabrication authority.

drawings/README.md

Drawing queue notes.

Drawings

Starter folder for lyre drawings. The first deliverable should be a dimensioned front elevation tied to string-schedule.csv, design.md, and wolfram-starter.wl.

Required first views:

- front elevation with all string paths, bridge, nut, yoke, pin row, and loaded grain direction;
- side section through the bridge/soundboard showing bridge height, break angle, bridge patch, and soundboard thickness;
- arm-to-yoke joinery detail with glue area, grain direction, and pin-row edge distance;
- bridge and nut slot detail after final gauges are selected.

Drawings are draft review artifacts until the schedule and load register are filled with selected string and prototype evidence.

assembly-manual.md

Shop-facing sequence.

Lyre Assembly Manual

Tools

Bandsaw or CNC router, drill press, brad-point bits, clamps, rasps/files, sanding blocks, tuner, calipers, straightedge, and stringing wrench.

Build Sequence

1. Make a full-size layout with string path, bridge, nut, yoke, and pin row.
2. Cut frame arms and crossbar oversize, keeping grain continuous through the loaded members.
3. Dry-fit the frame and mark the actual speaking lengths.
4. Cut and fit the soundboard or soundbox panel.
5. Glue the frame and soundboard with flat cauls and full clamp support.
6. Shape and polish the bridge and nut. Do not cut final slots until the gauge schedule is confirmed.
7. Drill pin pilot holes in scrap first, then in the yoke after torque testing.
8. Install strings gradually, moving across the set instead of loading one side fully before the other.
9. Tune in stages over 24 hours and record movement in validation.csv.

Safety

Wear eye protection when stringing. Stand out of the string plane during first pitch-up. If a frame member moves visibly, detune before inspecting.

validation.csv

Measurement and validation gates.

check_id	area	target	method	tolerance	measured	result	evidence	next_action
VAL-001	String tuning	C4 through E5 target pitches at A4=440	chromatic tuner after 24 hour settle	+/- 10 cents	TBD	TBD	string-schedule.csv plus pitch-up log	Adjust gauge, scale length, or bridge compensation

check_id	area	target	method	tolerance	measured	result	evidence	next_action
VAL-002	String safety	all strings below 70 percent estimated break	calculate from supplier density and break strength	<=70 percent	TBD	TBD	string-schedule.csv	Change gauge/material before full pitch
VAL-003	Frame deflection	no permanent yoke or arm movement under full tension	measure before stringing, half tension, full tension, and after 24 hours	<=0.030 in movement	TBD	TBD	bridge-soundboard-load-register.csv	Increase section, add lamination, or reduce total tension
VAL-004	Tuning pin hold	pins hold pitch through three retune cycles	mark pin angle and pitch drift on a scrap-block coupon	<=5 cents drift per hour	TBD	TBD	bridge-soundboard-load-register.csv	Revise pilot hole or pin spec
VAL-005	Ergonomics	comfortable lap/bench playing reach	player mockup photos and reach notes	no wrist strain	TBD	TBD	photo-shotlist.md plus mockup notes	Revise body width/string fan
VAL-006	Finish clearance	finish does not bind pins or string slots	dry fit after finish cure	free movement	TBD	TBD	assembly log	Mask or rework contact areas
VAL-007	Bridge and soundboard load	bridge/nut slots and soundboard tolerate first pitch-up	record bridge height, slot radius, break angle, top flatness, and deflection	no string cutting; no visible top crown or cracking	TBD	TBD	bridge-soundboard-load-register.csv	Rework bridge/nut or soundboard thickness before continuing

check_id	area	target	method	tolerance	measured	result	evidence	next_action
VAL-008	Joinery load path	arms, yoke, body, and pin row avoid short-grain failure	review full-size drawing or CAD detail before cutting final frame	loaded grain direction and pin edge distance explicitly shown	TBD	TBD	drawing-brief.md and bridge-soundboard-load-register.csv	Revise yoke/arm joint, pin row, or lamination strategy

supplier-rfq.md

Supplier RFQ template.

Supplier RFQ Draft

Hello,

I am sourcing parts for a 10-string prototype lyre. Please quote the following items or recommend substitutions that meet the same use case:

- 10 matching zither or harp tuning pins with recommended pilot-hole size.
- Nylon or fluorocarbon monofilament strings suitable for C4 through E5 at approximately 18-23 in speaking length.
- Optional matching tuning wrench.

Please include material specifications, breaking strength or rated tension, unit price, minimum order quantity, lead time, and shipping options.

Thank you.

visual-bom-brief.md

Visual BOM brief.

Visual BOM Brief

Create a one-page visual BOM with the assembled lyre at top, then callouts for frame hardwood, soundboard, tuning pins, strings, bridge, nut, adhesive, and finish. Use real photos or CAD renders when available. Mark any concept images as placeholders until replaced by shop or supplier images.

photo-shotlist.md

Documentation shot list.

Photo Shotlist

- Orthographic front view of full-size layout before cutting.
- Close-up of grain direction through arms and yoke.
- Pin pilot-hole test block with bit sizes labeled.
- Dry-fit frame before glue.
- Bridge and nut blank before and after slotting.
- First stringing with safety setup visible.
- Tuner readings for all 10 strings after 24 hour settle.

risks.md

Risk register.

Lyre Risks

Safety

- String breakage during first pitch-up can injure eyes or hands.
- Tuning pins may release suddenly if pilot holes are oversized.

Structural

- The yoke and arms carry the full string load and must avoid short-grain failure.
- Soundboard thickness is not validated; excessive bridge load may crown or crack the board.
- Arm-to-yoke joinery and pin-row edge distance are not yet backed by a reviewed drawing or proof-load coupon.

Acoustic

- Gauge schedule is not measured yet, so tone balance may be uneven.
- Bridge and nut slot quality can dominate sustain and string life.

Sourcing

- Live prices, availability, and lead times are not checked.
- Fishing-line substitutions are risky unless density and breaking strength are documented.

Readiness

Remain at prototype-planning scaffold status until the CAD layout, string schedule, bridge/soundboard load register, and first prototype validation rows are complete.

README.md

Repository overview.

Lyre

Root-mode build packet scaffold for a small plucked lyre, aligned with the instrument-maker packet shape.

Readiness

Prototype-planning scaffold. The packet now has a design basis, first-pass string assumptions, BOM, sourcing, cut list, validation plan, risks, photo plan, drawing brief, CAD notes, and Wolfram starter. It is not build-ready because no shop prototype, measured tuning data, reviewed CAD/DXF authority, sourceability check, or string-load proof has been completed.

Packet Contents

- explorer.html - studio overview linking the packet evidence, validation gates, print packet, Wolfram starter, CAD/drawing queues, and learn-to-play material.
- design.md - design intent, physics model, assumptions, and known gaps.
- bom.csv - first-pass parts and estimated costs.
- sourcing.csv - source specifications and search terms.
- cut-list.csv - rough and final stock plan.
- string-schedule.csv - C4-E5 target schedule with gauge/tension fields left measurement-required until supplier or sample-string data exists.
- bridge-soundboard-load-register.csv - bridge, soundboard, yoke, pin, and joinery load gates.
- visual-output-register.csv - records the reference image and review exports as non-authoritative visuals.
- validation.csv - tuning, load, ergonomic, and finish checks.
- assembly-manual.md - staged build notes for the first prototype.
- drawing-brief.md and drawings/README.md - drawing starter notes.
- cad/README.md - CAD starter notes.
- wolfram-starter.wl and wolfram/README.md - string model starter.

- risks.md - safety, structural, sourcing, and readiness risks.
- photo-shotlist.md - documentation image plan.

String-Scale Assumptions

Baseline: 10 strings, C4 through E5 diatonic, 21 in nominal speaking length, nylon or fluorocarbon monofilament, and a conservative prototype target below 70 percent of estimated breaking load. The exact gauge schedule remains a validation task in string-schedule.csv; empty tension fields are intentional and must not be treated as final load data.

Build Review Checklist

Before this packet moves beyond scaffold status, review the string chart against real supplier data. The lyre frame looks simple, but the crossbar and arms carry the entire string load continuously. The first review should confirm that no loaded member depends on short grain, that tuning pins have enough edge distance, and that the bridge/nut contact surfaces are smooth enough to avoid cutting monofilament strings.

CAD should start from a full-size front elevation. Keep the string paths as reference geometry, then derive the bridge, nut, yoke, and pin row from those paths. Do not hand-place the pin row independently from the string schedule.

Validation Boundary

This packet supports design review and prototype planning only. The following evidence is still required before claiming build-ready or validated status:

- Calculated gauge schedule using measured or supplier-provided density and breaking strength.
- Scrap-block tuning pin torque and creep test.
- CAD or full-size drawing that shows the string fan, yoke, bridge, nut, and loaded grain direction.
- First pitch-up log showing frame movement and tuning drift after 24 hours.
- Prototype photos that match the photo shotlist.

Shop Notes

Bring strings up to pitch in several passes rather than tuning one string to final pitch before the others. Record the before/after yoke distance, arm spread, and bridge height. If the frame moves visibly or a pin rotates under load, detune and revise the structure before continuing.

License

[CC BY 4.0](LICENSE) - see LICENSE for details.

wolfram-starter.wl

Local Wolfram starter.

```

(* Lyre string schedule starter. Values are first-pass assumptions. *)
ClearAll["Global`*"];
a4 = 440;
freqFromMidi[m_] := a4*2^((m - 69)/12);
stringTension[f_, scaleIn_, linearDensity_] := (2*scaleIn*f)^2*linearDensity;
percentBreak[scaleIn_, f_, densityLbIn3_, breakPsi_] :=
  densityLbIn3*4*scaleIn^2*f^2/(breakPsi*386.4)*100;

notes = <|"C4" -> 60, "D4" -> 62, "E4" -> 64, "F4" -> 65, "G4" -> 67,
  "A4" -> 69, "B4" -> 71, "C5" -> 72, "D5" -> 74, "E5" -> 76|>;
scaleLengthIn = 21;
nylonDensityLbIn3 = 0.04155;
nylonBreakPsi = 44600;

Dataset[
  KeyValueMap[
    <|"note" -> #1, "frequencyHz" -> N[freqFromMidi[#2]],
      "percentBreakIdeal" -> N[percentBreak[scaleLengthIn, freqFromMidi[#2],
        nylonDensityLbIn3, nylonBreakPsi]]|> &,
    notes
  ]
]

```